

Veritas AI Engineering Roadmap (July 7 - August End)

Week 1 – Foundation + Backend

Learn: FastAPI, PostgreSQL, Docker

Implement in Veritas: Build FastAPI backend, PostgreSQL integration, Docker Compose, POST /research API

Playlists: Patrick Loeber – FastAPI; freeCodeCamp PostgreSQL; TechWorld with Nana – Docker

Week 2 – Document Retrieval

Learn: APIs, arXiv, PubMed, PDF parsing, Chunking

Implement in Veritas: Retrieve papers, parse PDFs, chunk text

Playlists: James Briggs (Chunking); ArjanCodes (APIs)

Week 3 – Embeddings + Vector Search

Learn: Embeddings, Cosine Similarity, Sentence Transformers, Qdrant

Implement in Veritas: Generate embeddings and semantic search

Playlists: James Briggs – Embeddings & Vector DB

Week 4 – Hybrid Retrieval

Learn: BM25, Hybrid Search, RRF, Cross Encoder

Implement in Veritas: Dense + Sparse retrieval pipeline

Playlists: James Briggs; LangChain RAG

Week 5 – Knowledge Graph

Learn: Neo4j, Cypher, NER

Implement in Veritas: Extract claims and build KG

Playlists: Neo4j Official; spaCy NER

Week 6 – Verification ML

Learn: Transformers, NLI, Hugging Face, Fine-tuning

Implement in Veritas: Build contradiction classifier

Playlists: Andrej Karpathy; Hugging Face Course

Week 7 – Agents

Learn: ReAct, Tool Calling, LangGraph

Implement in Veritas: Planner + Retriever + Verifier + Synthesizer

Playlists: DeepLearning.AI; Cole Medin

Week 8 – Frontend + Production

Learn: Next.js, Tailwind, SSE, Deployment

Implement in Veritas: Dashboard, graph visualization, deploy

Playlists: Next.js Docs; TechWorld with Nana (GitHub Actions)

Daily Routine

2 hrs learning → 4 hrs implementation → 30 mins official docs → 30 mins GitHub commits → 30 mins revision.